

SONIA VELMURUGAN

RTL DESIGN VERIFICATION ENGINEER

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PROFESSIONAL SUMMARY

RTL Design Verification Engineer with hands-on experience developing SystemVerilog and UVM-based verification environments for AMBA APB, AMBA AHB, and Synchronous FIFO designs. Skilled in reusable UVM testbench development, constrained-random verification, functional coverage, code coverage, SystemVerilog Assertions (SVA), protocol compliance checking, and waveform-based debugging. Achieved 92% functional coverage for an AMBA APB verification project and 89% for AMBA AHB using Synopsys VCS and Questa Sim. Experienced across verification planning, testbench architecture, stimulus generation, coverage analysis, and debug.

TECHNICAL SKILLS

HDL & Verification Languages: Verilog HDL, SystemVerilog

Verification Methodologies: UVM, SystemVerilog Assertions (SVA), Constrained-Random Verification (CRV), Functional Coverage, Code Coverage

Protocols & Designs: AMBA APB, AMBA AHB, Synchronous FIFO

Verification Competencies: UVM Testbench Architecture, Reusable Testbench Development, Transaction-Based Verification, Verification Planning, Stimulus Generation, Coverage Analysis, Coverage Closure, Protocol Compliance Checking, Waveform Debugging

UVM Components: Sequence Item, Sequence, Sequencer, Driver, Monitor, Agent, Environment, Scoreboard, Virtual Interface, uvm_config_db

Simulation & EDA Tools: Synopsys VCS, Questa Sim, EDA Playground

Scripting & Operating Systems: Perl, Linux

DOMAIN-SPECIFIC VERIFICATION PROJECTS

UVM Verification of AMBA APB Slave Protocol

- Developed a reusable UVM-based verification environment for an AMBA APB Slave protocol, covering sequence item, sequence, sequencer, driver, monitor, agent, environment, and scoreboard components.
- Configured the UVM environment using uvm_config_db and connected the DUT through a virtual interface for transaction-level verification.
- Created constrained-random read/write stimulus to exercise protocol scenarios and improve functional coverage.
- Achieved 92% functional coverage with high code coverage using Synopsys VCS.
- Analyzed coverage gaps and protocol violations using waveform-based debugging and closed identified gaps through targeted directed tests.

Design and Verification of AMBA AHB Slave Protocol

- Designed an AMBA AHB Slave RTL in Verilog HDL and developed a SystemVerilog-based verification environment from scratch.
- Verified single-transfer, burst-transfer, read, and write operations for protocol-compliant behavior.
- Applied constrained-random scenario generation to exercise different transfer conditions and achieved 89% functional coverage using Synopsys VCS.
- Cross-checked simulation results in Questa Sim and performed waveform-based debug to identify and resolve verification failures.
- Performed coverage analysis to identify unverified scenarios and improve verification completeness.

Design and Verification of Synchronous FIFO

- Designed a Synchronous FIFO RTL in Verilog HDL with Full and Empty flag generation logic.
- Developed a task-based Verilog/SystemVerilog testbench to verify read/write operations, overflow, underflow, and boundary conditions.
- Verified Full/Empty flag behavior under constrained-random scenarios and evaluated functional and code coverage using Synopsys VCS.
- Analyzed FIFO pointer and flag behavior through waveform debugging to verify correct operation at boundary conditions.

B.E. ACADEMIC PROJECT

Smart Pole-Mounted Solar Panel System with Automated Disaster Protection and Web-Based Control Using IoT

- Developed an ESP32-based smart solar system with automated solar tracking, rain- and vibration-triggered panel folding, automatic wiper cleaning, web-based monitoring, EV charging support, and Twilio SMS disaster alerts.

EDUCATION

Advanced Diploma in ASIC Design – RTL Verification RV-Skills Design Center for Emerging Technologies	2026
Bachelor of Engineering – Electrical and Electronics Engineering University College of Engineering, Anna University CGPA: 7.8	2025
Higher Secondary Education CK School of Practical Knowledge Matriculation Higher Secondary School 86.44%	2021
SSLC Sri Kurinji Vidhyalaya Matriculation Higher Secondary School 82.2%	2019